fire and buried their dead. They did not practice agriculture and animal domestication.

Cro-Magnon was one of the most talked forms of modern human found from the rocks of Cro-Magnon, France and is considered as the ancestor of modern Europeans. They were not only adapted to various environmental conditions, but were also known for their cave paintings, figures on floors and walls.

Homo sapiens or modern human arose in Africa some 25,000 years ago and moved to other continents and developed into distinct races. They had a brain capacity of 1300 – 1600 cc. They started cultivating crops and domesticating animals.

Summary

Evolutionary Biology is the study of history of life forms on Earth which originated on Earth millions of years ago. How Earth originated, how life originated, what is the place of man in the universe are all general questions. This chapter deals with several theories to explain the life on Earth. Evidence from the fossil record and many other areas of biology like embryology, anatomy and molecular biology indicates a common ancestry.

The theories advanced by Lamarck, Darwin, Hugo de Vries explained the intricate evolutionary process. Geological time scale with different eras, periods and epochs gives an idea about the dominant species in those days. The mathematical distribution of gene and genotype frequencies remains constant in a small population was contributed by Hardy and Weinberg in 1608. Natural Selection and gene pool are the important factors those affect Hardy Weinberg equilibrium.

Human evolution states that humans developed from primates or ape like ancestors. The emergence of Homo sapiens as a distinct species from apes and placental mammals in brain size, eating habit and other behavior proves that 'Ontogeny recapitulates Phylogeny'.

Evaluation

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1) The first life on earth originated
   a) in air
   b) on land
   c) in water
   d) on mountain

2) Who published the book "Origin of species by Natural Selection" in 1859?
   a) Charles Darwin b) Lamarck
   c) Weismann d) Hugo de Vries

3) Which of the following was the contribution of Hugo de Vries?
   a) Theory of mutation
   b) Theory of natural Selection
   c) Theory of inheritance of acquired characters
   d) Germplasm theory

4) The wings of birds and butterflies is an example of
   a) Adaptive radiation
   b) convergent evolution
   c) divergent evolution
   d) variation

5) The phenomenon of " Industrial Melanism" demonstrates
   a) Natural selection
   b) induced mutation
   c) reproductive isolation
   d) geographical isolation

6) Darwin's finches are an excellent example of
   a) connecting links b) seasonal migration
   c) adaptive radiation d) parasitism

7) Who proposed the Germplasm theory?
   a) Darwin **b) August Weismann**
   c) Lamarck d) Alfred Wallace

8) The age of fossils can be determined by
   a) electron microscope
   b) weighing the fossils
   c) carbon dating
   d) analysis of bones

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